Optimized Mueller polarimeter with liquid crystals

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Received September 9, 2002

We demonstrate a Mueller polarimeter in which the polarization-state generator and analyzer are both composed of a linear polarizer and two liquid-crystal variable retarders. The polarimeter is designed to optimize the accuracy of the final results by minimization of the condition numbers of the modulation and analysis matrices. The polarimeter calibration, a difficult task by conventional procedures, is achieved easily by use of the eigenvalue method of Compain *et al.* [Appl. Opt. **38,** 3490 (1999)]. The overall polarimeter performance is tested with a linear polarizer at various angles and a compensator at various retardations. © 2003 Optical Society of America

OCIS codes: 120.5410, 230.5440, 260.5430.

A complete and general polarimetric characterization of any sample is given by its Mueller matrix **M**, defined by the linear relationship $\mathbf{S}_{\text{out}} = \mathbf{M}\mathbf{S}_{\text{in}}$, where $\mathbf{S}_{\text{in}}$ and $\mathbf{S}_{\text{out}}$ are the Stokes vectors of the incoming and outgoing light.

A Mueller matrix polarimeter is always composed of a polarization-state generator (PSG), which generates at least four linearly independent basis states $\mathbf{S}_{\text{in}}$, and a polarization-state analyzer (PSA), which analyzes $\mathbf{S}_{\text{out}}$ by measuring its projections over at least another four linearly independent basis states. Here, we consider a single-channel system with a PSG generating exactly four Stokes vectors $\mathbf{S}_{\text{in}}$, which are the column vectors of the modulation matrix, **W**. Similarly, the PSA is assumed to project $\mathbf{S}_{\text{out}}$ over exactly four Stokes vectors, which are the row vectors of the analyzer matrix, **A**. Then, a complete set of 16 measurements on a sample characterized by a Mueller matrix **M** can be written in matrix form:

$$\mathbf{B} = \mathbf{A}\mathbf{M}\mathbf{W}, \tag{1}$$

and **M** can be extracted readily from the raw data matrix, **B**, provided that **A** and **W** are known, i.e., if the system is calibrated.

Light polarization can be modulated and analyzed by a variety of approaches: rotating retardation plates,[1–3] rotating compensators,[4] Pockels cells,[5] photoelastic modulators,[6–8] or liquid-crystal (LC) variable retarders.[9,10] In this Letter we present an optimized device based on the last-named approach that has some limitations, in particular, the accessible spectral range, but also significant advantages, such as the absence of moving parts or high driving voltages.

The polarimeter geometry is shown in Fig. 1. The PSG is composed of a linear polarizer, P, followed by two nematic LC cells. The LC axes with variable indices are placed at azimuthal angles θ_1 and θ_2 with respect to the transmission axis of the polarizer. The arrangement of the PSA is similar, with its elements in reverse order. The transmission axes of the linear polarizer and analyzer are not necessarily aligned.

It has been shown that the error propagation from the raw data, **B**, to the matrix, **M**, is optimized when **A** and **W** are as close as possible to unitary matrices, i.e., when their condition numbers are minimized.[8,11–13] The definition of the condition number $\kappa(\mathbf{X})$ of a square matrix **X** is not unique.[11,13] We retain the most usual definition, which is also the most directly related to noise propagation: $\kappa(\mathbf{X})$ is the ratio of the largest over the smallest of the singular values of **X**.[11]

The PSG is operated by defining sequentially four pairs of retardation values (δ_1, δ_2) for the two LCs. Calculations analogous to those outlined in Refs. 11 and 13 show that $\kappa(\mathbf{W})$ exhibits several global minima, all equal to $\sqrt{3}$. Some of these minima are reached for a retardation sequence of the form $(\delta_1, \delta_2) = (\Delta_1, \Delta_1)$, (Δ_1, Δ_2), (Δ_2, Δ_1), (Δ_2, Δ_2), with

$$\Delta_1 = 315^\circ + p90^\circ, \qquad \Delta_2 = 135^\circ + p90^\circ, \tag{2}$$

where p is the same integer for Δ_1 and Δ_2, and for LC orientations given by

$$\theta_1 = \epsilon 27.4^\circ + q90^\circ, \qquad \theta_2 = \epsilon 72.4^\circ + r90^\circ, \tag{3}$$

where $\epsilon = \pm 1$ has the same value for θ_1 and θ_2 and q and r are any integers, not necessarily equal. The same results clearly apply to the optimization of the PSA.

We now address the key issue of polarimeter calibration. The actual basis states of both the PSG and

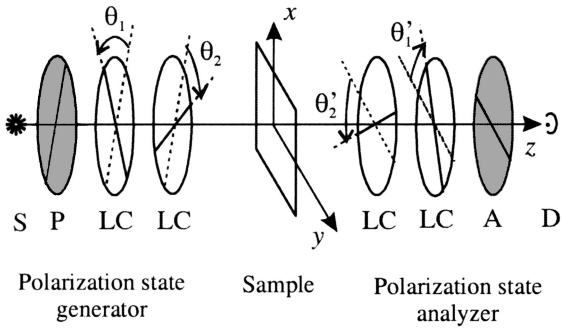


Fig. 1. Schematic of the polarimeter: S, source; P, linear polarizer; A, analyzer; D, detector. The variable-index axes of the LCs, shown as solid lines, are oriented at θ_i (θ_i') azimuthal angles with respect to the transmission axis of the linear polarizer (the analyzer). The xy axes in the transverse plane are defined by the calibration procedure.

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the PSA depend quite sensitively on many independent parameters. As a result, it is difficult to calibrate a LC-based polarimeter accurately by the usual methods. This difficulty is even more pronounced for the optimized PSG and PSA, which involve elliptical basis states, than for more intuitive LC-based configurations, for which the target basis states, typically linear and circular, are easier to characterize.

These difficulties are solved easily by the eigenvalue calibration method, described in detail in Ref. 14. Briefly, the idea is to determine both $\mathbf{A}$ and $\mathbf{W}$ from the raw data matrices $\mathbf{B}_i$ taken with a set of calibration samples consisting of suitably chosen dichroic retarders (DRs). The Mueller matrix, $\mathbf{M}(\tau, \Psi, \Delta)$, of a DR in its principal axes is a reflection matrix, with the following nonvanishing elements:

$$\mathbf{M}_{11} = \mathbf{M}_{22} = \tau\,, \qquad \mathbf{M}_{12} = \mathbf{M}_{21} = -\tau \cos 2\Psi\,,$$
$$\mathbf{M}_{33} = \mathbf{M}_{44} = \tau \sin 2\Psi \cos\Delta\,,$$
$$\mathbf{M}_{43} = -\mathbf{M}_{34} = \tau \sin 2\Psi \sin\Delta\,, \tag{4}$$

where Ψ and Δ are the usual ellipsometric angles and τ is proportional to the DR transmission. The Mueller matrix, $\mathbf{M}_i$, of calibration sample i set at an azimuthal angle ϕ_i is then

$$\mathbf{M}_i = \mathbf{R}(-\phi_i)\mathbf{M}(\tau_i, \Psi_i, \Delta_i)\mathbf{R}(\phi_i)\,, \tag{5}$$

where $\mathbf{R}(\phi)$ is the matrix of a rotation by an angle ϕ about the z axis. Clearly, the eigenvalues of $\mathbf{M}_i$ are independent of ϕ_i and equal to $[2\tau \sin^2\Psi,\ 2\tau \cos^2\Psi,\ \tau \sin 2\Psi \exp(\pm i\Delta)]$.

The essential steps of the procedure are the following: (a) A first measurement is carried out without any sample, yielding $\mathbf{B}_0 = \mathbf{A}\mathbf{W}$; (b) next, measurements are made with the DRs in place. From the raw data matrices $\mathbf{B}_i$ we calculate the matrices $\mathbf{C}_i = (\mathbf{B}_0)^{-1}\mathbf{B}_i = \mathbf{W}^{-1}\mathbf{M}_i\mathbf{W}$, which clearly have the same eigenvalues as $\mathbf{M}_i$ and thus provide the actual values of τ_i, Ψ_i, and Δ_i independently of the DR orientation ϕ_i; (c) $\mathbf{W}$ is determined as the unique solution of the set of equations $\mathbf{M}_i\mathbf{W} - \mathbf{W}\mathbf{C}_i = 0$. To this end, a linear operator $\mathbf{K}$ is built in the 16-dimensional space of 4×4 real matrices so that its only eigenvector associated with a zero eigenvalue is $\mathbf{W}$. In fact, the definition of $\mathbf{K}$ involves both the experimentally known matrices $\mathbf{C}_i$ and the values of $(\tau_i, \Psi_i, \Delta_i)$ calculated in step (b), as well as the unknown angles ϕ_i, which are determined together with $\mathbf{W}$ by the requirement that one of the eigenvalues of $\mathbf{K}$ actually vanish. (d) $\mathbf{A}$ is eventually calculated as $\mathbf{A} = \mathbf{B}_0\mathbf{W}^{-1}$.

A convenient set of calibration samples contains a linear polarizer set at two orientations ($\phi_1 = 0$ by definition, and $\phi_2 \approx 90°$) and a birefringent plate providing a retardation value Δ_3 anywhere from 80° to 140°, set at $\phi_3 \approx 30°$. As the actual values of ϕ_2, ϕ_3, and Δ_3 are determined accurately during the calibration itself, no precise positioning systems or specific optical elements (such as exact quarter- or half-wave plates) are needed.

We also emphasize that many effects that can significantly modify $\mathbf{A}$ and $\mathbf{W}$ with respect to their target values (e.g., inaccuracies in the LC orientations and retardations, interference effects, beam divergence, strain-induced birefringence) are automatically taken into account, without the need for any modeling, as long as these effects are static and do not alter the linearity of the intensity measurements. This is a major issue for a system involving as many independent parameters as the polarimeter described here. As a matter of fact, even when the PSG and PSA are set far from autocollimation, or one or several lenses are inserted or removed, the measurement accuracy described below is always fully recovered, provided that the system is calibrated in its actual configuration. The only critical point is that the calibration samples must be as close as possible to an ideal DR. This is usually no problem for linear polarizers, but it may be more of a concern for retarders.

The LCs used in this work were PLM-20 LCs from Jenoptik, without antireflection coatings. The mechanical mounts of the PSG and PSA were very simple and rugged, without precision rotation stages. The light source was a 5-mW He–Ne laser. With a LC temperature regulated at 35 °C the retardation switching times were typically 60 ms, resulting in a total acquisition time of ~1 s for a complete Mueller matrix measurement. The values of $\kappa(\mathbf{A})$ and $\kappa(\mathbf{W})$ provided by the calibration procedure were typically from 1.9 to 2.1, reasonably close to the theoretical

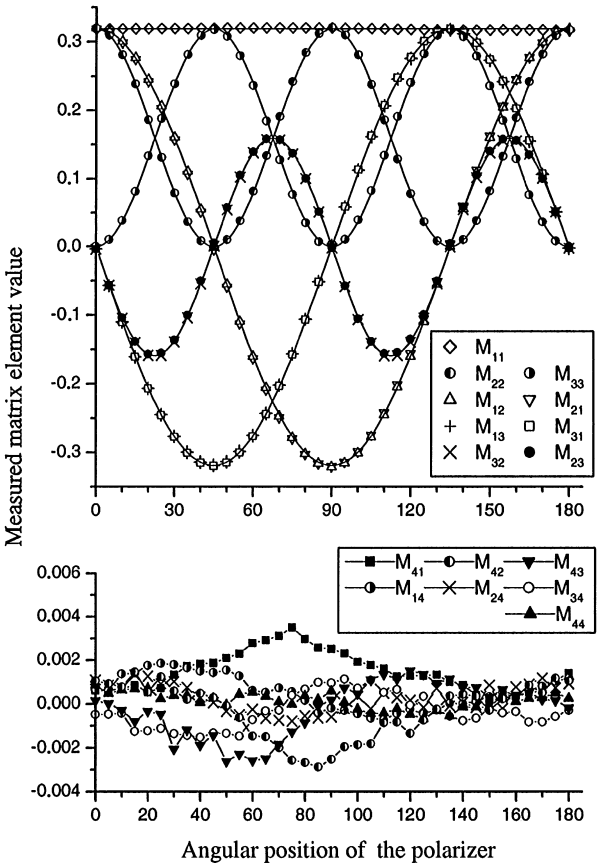


Fig. 2. Mueller matrices measured with a linear polarizer set at various orientations in the xy plane. Top, nonvanishing elements (symbols) and single parameter fits (solid curves); bottom, elements that are expected to vanish.

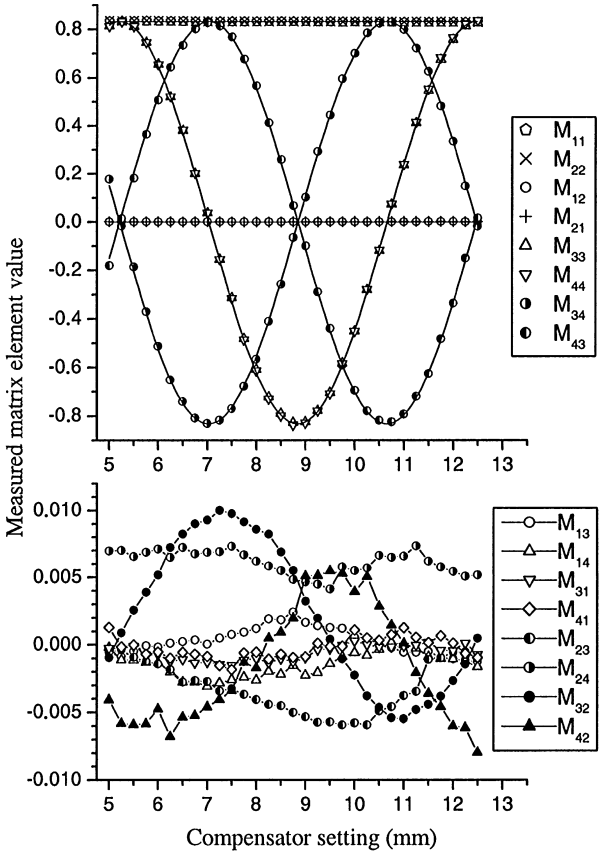


Fig. 3. Mueller matrices measured with a compensator, aligned with the x axis, for different settings of the micrometric screw controlling the compensator retardation. Top, nonvanishing elements (symbols) and fits (solid curves); bottom, other elements, which are expected to vanish.

minimum 1.73, and were reproducible to within 0.005 for a given system configuration.

To evaluate the overall system performance we made two series of Mueller matrix measurements, with two easily characterizable test samples, namely, a linear polarizer (a dichroic polarizer from Melles–Griot), measured at various orientations θ_p in the xy plane, and a high-quality Babinet–Soleil quartz compensator (Optique Fichou), aligned with the x axis and measured at various settings of the micrometric screw controlling its retardation.

The results obtained with the polarizer are summarized in Fig. 2. As the polarizer was the same as that used for calibration and was left on the same rotation stage, the absolute value of θ_p was known with the accuracy of the angle reading, $\sim 0.3°$. The top graph shows the results obtained for the elements of the 3×3 upper diagonal block. All the data have been fitted with a single parameter, the polarizer transmission, with an optimal value of 0.6378. This value agrees quite well with that determined from the nonvanishing eigenvalues of the polarizer **C** matrices (0.636 ± 0.002). The bottom graph shows that the other elements, expected to vanish, are actually quite small. The rms deviation between the measured and fitted values (including the vanishing elements) for the whole series of measurements is 0.0016, corresponding to 0.005 for normalized Mueller matrices.

The data taken with the compensator aligned along the x axis are summarized on Fig. 3. The nonvanishing elements, shown in the top graph of Fig. 3, have been fitted according to Eqs. (4) with three parameters, namely, the compensator transmission and the two constants describing the linear relationship between the retardation, Δ, and the setting of the compensator micrometric screw. The rms deviation between measured and fitted data is 0.009, corresponding to 0.011 for normalized matrices. This somewhat larger rms deviation may be due to the fact that, like achromatic wave plates,[15] compensators are more difficult to make perfect than polarizers. However, the variation of retardation Δ (as deduced directly from the measured elements) with the screw setting is found to be linear, with a rms deviation of 0.13°, equivalent to $\lambda/2700$, which is even better than the manufacturer specification.

To summarize, we have developed an optimized configuration of a Mueller polarimeter based on liquid-crystal variable retarders, and we demonstrated that with a suitable and easy-to-use calibration method this instrument can be built in a rugged fashion and yet provide quite a satisfactory measurement accuracy.

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